

Dahyoun Lee

PhD Student (Integrated MS/PhD)
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EDUCATION

Mar. 2024 ~ Present	Yonsei University Department of Medical Systems Informatics <i>Advisor: Byunghoon Kim, Sengchan You</i> <i>Ph.D. Student</i> GPA: 4.13 / 4.3	Seoul, Korea
Mar. 2019 ~ Feb. 2024	Ajou University Department of Biological Science / Artificial Intelligence in Medicine <i>B.S. in Biological Science</i> GPA: 4.11 / 4.5	Suwon, Korea

RESEARCH INTERESTS

- Large language model
- Multimodal learning
- Neuroscience
- Brain network
- Psychiatry

RESEARCH EXPERIENCES

- Research Student at Neurostat Lab(prof.B.Park), Department of Biomedical Informatics, Ajou University School of Medicine, Korea (Mar. 2023 ~ Dec. 2023) / Brain image preprocessing, Visual inspection assistant, Developing dementia classification models &
- Research Student at Circadian clock Lab(prof. E.Kim), Department of Brain Science, Ajou University School of Medicine, Korea (May. 2022 ~ Dec. 2022) / Experiments to explore the Drosophila circadian clock genes

AWARDS AND HONORS

- Master's Student Research Fellowship, National Research Foundation of Korea(NRF), Ministry of Education, Korea (Jul. 2024 ~ Jun. 2025)
- [Artificial Intelligence in Medicine] Academic Excellence Scholarship & Self-development

Scholarship, Industry-Academic Cooperation Foundation, Ajou University College of Medicine, Korea (Jan. 2023)

- Leader Scholarship (Student representative), The department of Biological Science, Ajou University., Korea (Dec. 2022)

SKILLS AND TECHNIQUES

- Python
- Pytorch and Tensorflow
- MATLAB for Brain imaging processing and Machine learning
- Statistics (R)
- Bioinformatics (Biopython, Linux)

PUBLICATIONS (SCIE/ESCI)

1. Pae Sun Suh, Dahyoun Lee, Chang-Bae Bang et al, "Predicting molecular types of adult-type diffuse gliomas based on MRI reports with large language models", *EUROPEAN RADIOLOGY*, (2025), <https://doi.org/10.1007/s00330-025-12211-x>

PUBLICATIONS (PREPRINT)

- Junhyeok Lee; Yujin Oh; Dahyoun Lee; Hyon Keun Joh; Chul-Ho Sohn; Sung Hyun Baik; Cheol Kyu Jung; Jung Hyun Park; Kyu Sung Choi; Byung-Hoon Kim et al. (2024). Improving Factuality of 3D Brain MRI Report Generation with Paired Image-domain Retrieval and Text-domain Augmentation. arXiv preprint. <https://doi.org/10.48550/arXiv.2411.15490>

PROJECTS

- Multi-task Multi-modal Brain Signal Continuous Measurement-based Young Adult ADHD Screening and Cognitive Function Assessment AI Algorithm Product Development (Bio&Medical Technology Development Program), National Research Foundation of Korea(NRF), Ministry of Education, Korea / Research Assistant (Apr. 2025 ~ Present)
- Construction of cohort-based brain imaging AI learning data for cognitive aging study, Korea Disease Control and Prevention Agency, KDCA, Korea / Research Assistant (Apr. 2025 ~ Present)
- Representation Learning of Brain Connectivity with Graph Neural Networks for Investigation of Biomarkers in Schizophrenia (Basic Science Research Program), National Research Foundation of Korea(NRF), Ministry of Education, Korea / Researcher (Data Management & Report Writing) (Mar. 2024 ~ May. 2025)